DAY-4

Subnetting Plan Assignment

Q1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Answer:

Subnet	Network address	Subnet mask	Broadcast address	Usable host range
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.63	192.168.10.1 - 192.168.10.62
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.127	192.168.10.65 - 192.168.10.126
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.191	192.168.10.129 - 192.168.10.190
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.255	192.168.10.193 - 192.168.10.254

All four subnets use the same mask, 255.255.255.192 (/26), and each has 62 usable hosts.

Q2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Answer:

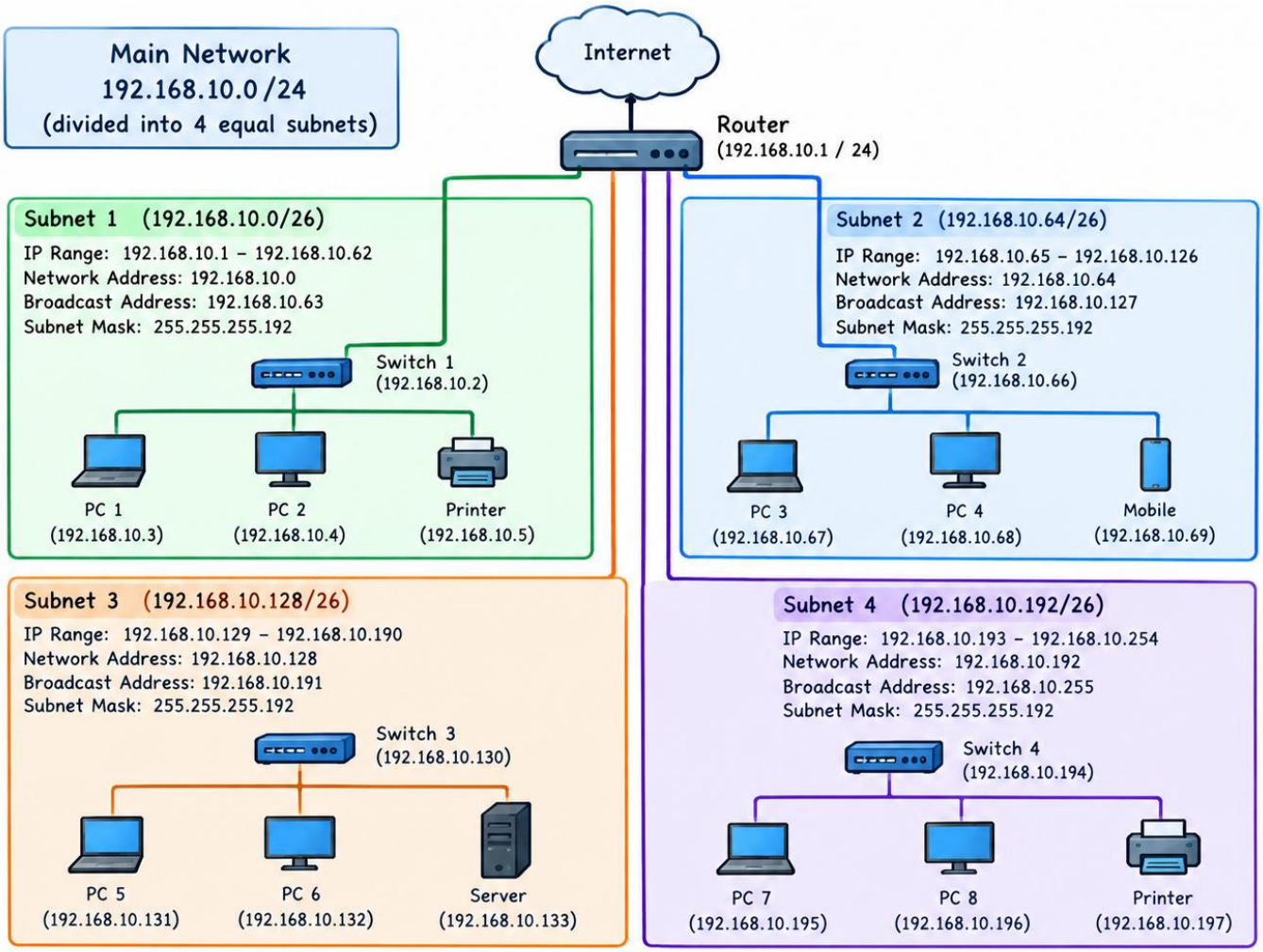
I chose Subnet 1 (192.168.10.0/26). Its network address is 192.168.10.0 and its broadcast address is 192.168.10.63, so these two are not used. All devices use addresses from 192.168.10.1 to 192.168.10.62.

Device	IP address	Subnet mask	Default gateway
Router (gateway interface)	192.168.10.1	255.255.255.192	-
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1
Mobile	192.168.10.5	255.255.255.192	192.168.10.1

The router takes the first usable address (192.168.10.1) and acts as the default gateway. The remaining addresses 192.168.10.6 to 192.168.10.62 are free for future devices.

Q3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Answer:



Subnet	Network Address	Broadcast Address	Subnet Mask	Valid Host Range	Total IPs	Usable Hosts
Subnet 1	192.168.10.0	192.168.10.63	255.255.255.192	192.168.10.1 – 192.168.10.62	64	62
Subnet 2	192.168.10.64	192.168.10.127	255.255.255.192	192.168.10.65 – 192.168.10.126	64	62
Subnet 3	192.168.10.128	192.168.10.191	255.255.255.192	192.168.10.129 – 192.168.10.190	64	62
Subnet 4	192.168.10.192	192.168.10.255	255.255.255.192	192.168.10.193 – 192.168.10.254	64	62

The router has one interface in every subnet. The four devices of Subnet 1 are connected to a switch, which is connected to the router interface Gi0/0 (192.168.10.1).

Q4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Answer:

Think of the network 192.168.10.0/24 as a big pizza with 256 slices (addresses). We want to cut it into 4 equal parts, so every part (subnet) gets 64 slices. Here is how to do it:

Step 1: Understand what /24 means. An IP address has 32 bits. /24 means the first 24 bits are for the network and the last 8 bits are for hosts. So 2^8 = 256 addresses are available in total, and the mask is 255.255.255.0.

Step 2: Find how many bits to borrow. We need 4 subnets. The number of subnets is 2^n, so 2^2 = 4. This means we borrow 2 bits from the host part.

Step 3: Find the new prefix and mask. New prefix = 24 + 2 = /26. In binary the mask is 11111111.11111111.11111111.11000000. The last octet is 128 + 64 = 192, so the mask is 255.255.255.192.

Step 4: Find the hosts per subnet. Host bits left = 32 - 26 = 6. Addresses per subnet = 2^6 = 64. Two of them are reserved (the first is the network address and the last is the broadcast address), so usable hosts = 64 - 2 = 62.

Step 5: Find the block size. Block size = 256 - 192 = 64. This is the jump between subnets, so the subnets start at 0, 64, 128 and 192.

Step 6: Write each subnet. Network address = the starting number. Broadcast address = the next starting number minus 1. Usable hosts = network + 1 to broadcast - 1. For example, Subnet 1: network 192.168.10.0, broadcast 192.168.10.63, hosts 192.168.10.1 to 192.168.10.62.

Step 7: Assign IP addresses. I picked Subnet 1. I gave the router 192.168.10.1 (the gateway), Laptop 1 192.168.10.2, Laptop 2 192.168.10.3, the printer 192.168.10.4 and the mobile 192.168.10.5. I did not use 192.168.10.0 (network) or 192.168.10.63 (broadcast).

Step 8: Check the answer. A device is in the subnet if its IP AND the mask gives the network address. For the mobile, 192.168.10.5 AND 255.255.255.192: the last octet 5 = 00000101 and 192 = 11000000, so 00000101 AND 11000000 = 00000000 = 0. The result is 192.168.10.0, so the mobile is correctly in Subnet 1.